# Virtual Sub-States and the Reversible Layer of Loschmidt's Paradox

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**Abstract**

Loschmidt's reversibility objection observes that the microscopic equations of motion are invariant under time reversal, while macroscopic processes are not, and asks how the second law can follow from time-symmetric mechanics. We argue that the objection conflates two strata of a system's description that are governed by different logics, and that separating them dissolves the paradox without recourse to probabilistic improbability, coarse-graining, or special boundary conditions. The first stratum is *configurational*: it consists of the instantaneous microstate together with all decompositions of that microstate that recover it under a fixed composition law. On this stratum, evolution is genuinely reversible, and there exists a large space of *virtual sub-states*—intermediate decompositions whose components may fall outside the physically admissible region while their composite remains admissible. The second stratum is *enumerative*: it consists of the monotone count of completed dynamical events the system has undergone. We prove that this count cannot decrease, because any operation that would decrement it is itself an event that increments it; that the configurational reversibility and the enumerative monotonicity are logically independent; and, centrally, that virtual sub-states are available only for the backward completion of an already-fixed endpoint and never for forward construction—an asymmetry we make precise as a theorem about composition contexts. Velocity reversal acts on the configurational stratum and leaves the enumerative stratum untouched; the second law is the monotonicity of the latter and is therefore immune to the former. We then exhibit two independent physical realisations of virtual sub-states—the off-shell internal lines of quantum field theory, where path opacity is the statement that only the sum over diagrams is observable, and the phase-coherent sub-components of a Fourier-transform mass-spectrometry transient, where a measured fraction of the decomposition lies off the physical manifold—and show that the nuclear spin echo is a controlled laboratory test of the reversibility assumption whose seventy-year verdict is an engineering-independent floor. The framework is presented as mathematics with physical interpretation sequestered to remarks; every theorem is proved from stated definitions.

**Keywords:** Loschmidt's paradox; arrow of time; irreversibility; reversible dynamics; virtual sub-states; off-shell states; path opacity; spin echo; partition count; second law of thermodynamics.

## Contents

# 1 Introduction

## 1.1 The objection and its persistence

In 1876 Josef Loschmidt raised an objection to Boltzmann's programme of deriving the second law of thermodynamics from the mechanics of atoms (Loschmidt, 1876). The microscopic equations of motion—in their canonical form, Hamilton's equations—possess a symmetry: if one takes any solution and simultaneously reverses the sign of time and the signs of all momenta, the result is again a solution. The equations do not distinguish future from past. Yet macroscopic processes are flagrantly asymmetric in time: heat flows from hot to cold and never the reverse, a gas released into a vacuum disperses and never spontaneously re-concentrates. The objection, in its sharpest form, is this. Let a gas evolve from a low-entropy macrostate at time $t_0$ to a high-entropy macrostate at time $T$. Reverse every molecular velocity at $T$. The reversed state is a legitimate solution of the same equations, and it will retrace the expansion in reverse, the gas re-collecting itself, entropy decreasing. The laws permit it. Why is it never seen?

Boltzmann's reply was statistical (Boltzmann, 1877): the reversed evolution is not impossible, only overwhelmingly improbable, because the microstates leading to entropy decrease occupy a vanishing fraction of the high-entropy macrostate. Over the following century this reply was refined by Gibbs's coarse-graining (Gibbs, 1902), the information-theoretic reformulation (Jaynes, 1957), the cosmological-boundary account (Penrose, 1989), and environmental decoherence (Zurek, 2003); the conceptual literature is large (Ehrenfest and Ehrenfest, 1959; Lebowitz, 1993; Price, 1996; Reichenbach, 1956; Zeh, 2007). These accounts differ in emphasis but share a single structural commitment, which we will call the *reversibility presupposition*: that time reversal is a well-defined, in-principle-achievable operation whose macroscopic consequences merely fail to be realised, for contingent reasons such as improbability, ignorance, openness, or special initial conditions. Each account then explains why the reversed evolutions, though available, are not seen.

## 1.2 What this paper does differently

We do not accept the framing and then explain away its consequences. We argue that Loschmidt's intuition silently identifies two strata of a system's description that obey different logics, and that the appearance of paradox is entirely an artefact of treating them as one. The two strata are these.

- The *configurational stratum*: the instantaneous microstate of the system, together with the entire space of ways of decomposing that microstate into parts that recompose to it under a fixed composition law. On this stratum the dynamics is genuinely reversible; nothing in what follows disputes that. We will moreover show that this stratum is far larger than the microstate alone, because it contains a rich family of *virtual sub-states*: decompositions whose individual components need not be physically admissible, provided their composite is.
- The *enumerative stratum*: the integer count of completed dynamical events—closed oscillation cycles, committed interactions, recorded distinctions—that the system has undergone since some origin. This count is the operational content of elapsed time, and it cannot decrease.

The thesis of the paper is that velocity reversal is an operation on the configurational stratum that leaves the enumerative stratum untouched, that the second law is the monotonicity of the enumerative stratum, and that the two are logically independent, so the former cannot affect the latter. The reason velocity reversal feels as though it should reverse *everything* is the same reason virtual sub-states feel paradoxical: both rely on the freedom of the configurational stratum, and both are silent about the count. We make this precise through a structural asymmetry—virtual sub-states are available for the backward completion of a fixed endpoint but never for forward construction (Section 6)—which is the configurational shadow of the enumerative monotonicity.

Two features distinguish the resulting resolution from the standard ones. First, it is not probabilistic: nothing here depends on the reversed microstate being rare. Second, virtual sub-states are not a metaphor but an identifiable structure with two independent physical realisations, developed in Sections 8 and 9, and the central experimental signature—the nuclear spin echo—is a controlled laboratory test of the reversibility presupposition itself (Section 10), with a definite and repeatedly confirmed verdict.

## 1.3 Organisation

Section 2 fixes the mathematical setting: bounded measure-preserving dynamics, the microstate, the macroscopic observable, and the event count, with the three kept rigorously distinct. Section 3 establishes the monotonicity of the count and the incoherence of its decrement. Section 4 defines the configurational stratum and proves its reversibility. Section 5 introduces virtual sub-states and proves their existence and abundance. Section 6 proves the forward/backward asymmetry that is the keystone of the resolution. Section 7 assembles the resolution of Loschmidt's paradox and contrasts it with the standard accounts. Section 8 develops the quantum-field-theoretic realisation of virtual sub-states; Section 9 the mass-spectrometric one; Section 10 the spin-echo realisation and its experimental hierarchy. Section 11 answers objections; Section 12 concludes. Physical interpretation is sequestered to Remarks throughout; the logical spine is carried entirely by the formal environments.

# 2 The Mathematical Setting

## 2.1 Bounded measure-preserving dynamics

We work with a dynamical system in the standard sense of ergodic theory (Arnol'd, 1989; Walters, 1982).

**Definition 2.1** (Dynamical system)**.** A *dynamical system* is a triple $(\Omega, \mu, \phi_t)$, where $\Omega$ is a set (the *phase space*) carrying a $\sigma$-algebra of measurable sets, $\mu$ is a measure on $\Omega$, and $\phi_t \colon \Omega \to \Omega$, $t \in \mathbb{R}$, is a one-parameter group of measurable maps (the *flow*) with $\phi_0 = \mathrm{id}$ and $\phi_{t+s} = \phi_t \circ \phi_s$. A point $x \in \Omega$ is a *microstate*.

**Definition 2.2** (Bounded, measure-preserving)**.** The system $(\Omega, \mu, \phi_t)$ is *bounded* if $0 < \mu(\Omega) < \infty$, and *measure-preserving* if $\mu(\phi_t^{-1}(A)) = \mu(A)$ for every measurable $A$ and every $t$.

The two conditions are the abstract residue of two ubiquitous physical facts: that a persistent system occupies a bounded region of its phase space (an electron does not leave the atom, a gas molecule does not leave the box, a star does not leave the galaxy), and that Hamiltonian flow preserves phase-space volume (Liouville, 1838). We take both as given; the development below uses only their measure-theoretic content.

**Theorem 2.3** (Recurrence)**.** *Let $(\Omega, \mu, \phi_t)$ be bounded and measure-preserving. For every measurable $A \subseteq \Omega$ with $\mu(A) > 0$ and every $T > 0$, there exists an integer $n \geq 1$ with $\mu(A \cap \phi_{nT}^{-1}(A)) > 0$. Consequently $\mu$-almost every point of $A$ returns to $A$ under iteration of $\phi_T$.*

*Proof.* This is the Poincaré recurrence theorem (Poincaré, 1890); we recall the proof because the structure recurs below. Fix $T > 0$ and write $f := \phi_T$. The sets $A, f^{-1}(A), f^{-2}(A), \ldots$ all have measure $\mu(A)$ by measure preservation. If the conclusion failed, then $\mu(A \cap f^{-n}(A)) = 0$ for all $n \geq 1$, and an elementary computation shows the sets $\{f^{-k}(A)\}_{k \geq 0}$ are pairwise disjoint modulo null sets. Then $\mu(\Omega) \geq \sum_{k \geq 0} \mu(f^{-k}(A)) = \sum_{k \geq 0} \mu(A) = \infty$, contradicting boundedness. Hence some $n \geq 1$ has $\mu(A \cap f^{-n}(A)) > 0$. The pointwise statement follows by the standard argument (Walters, 1982). □

*Remark* 2.4. Recurrence is the first appearance of the tension Loschmidt's objection exploits. A bounded measure-preserving system returns arbitrarily close to any prior microstate, infinitely often. If returning the microstate were the same as returning the system to its past, the second law would be violated at every recurrence. We will see (Theorem 4.6) that it is not.

## 2.2 Time reversal of the flow

**Definition 2.5** (Time-reversal involution)**.** A measurable involution $R \colon \Omega \to \Omega$ (so $R \circ R = \mathrm{id}$) is a *time reversal* for the flow if

$$\phi_{-t} = R \circ \phi_t \circ R \qquad \text{for all } t \in \mathbb{R}. \tag{1}$$

For a Hamiltonian system with momenta $p$ and positions $q$, the standard choice is $R(q, p) = (q, -p)$, and (1) holds whenever the Hamiltonian is even in $p$.

*Remark* 2.6 (The hinge). Equation (1) is an identity among *solutions* of the equations of motion: it says that a certain transformation carries one solution to another. It is silent about whether there exists a *physical operation*, performable on a system in a laboratory, whose execution carries the system from the first solution to the second. The whole of Loschmidt's paradox turns on the slide from the former to the latter, and the whole of our resolution is the refusal to make that slide. The configurational stratum is where (1) lives; the enumerative stratum is where physical operations live; they are different strata.

### 2.3 Three quantities, kept distinct

The conceptual core of the setting is a separation that Loschmidt's intuition collapses. For a system of many constituents we distinguish three quantities.

**Definition 2.7** (Microstate, observable, count)**.** Let $(\Omega, \mu, \phi_t)$ be a bounded measure-preserving system.

(i) The *microstate* $\mu(t) := \phi_t(x_0) \in \Omega$ is the complete instantaneous specification of the system started from $x_0$.

(ii) An *observable* is a measurable map $\mathcal{O}\colon \Omega \to E$ into some value space $E$; the *observable value* is $\mathcal{O}(\mu(t))$. An observable is *coarse* if it is many-to-one on a set of positive measure, so that distinct microstates share an observable value.

(iii) The *event count* $N(t) \in \mathbb{Z}_{\geq 0}$ is the number of completed dynamical events the system has undergone in $[0, t]$, where an *event* is the completion of a distinguished, recurrent dynamical cycle (made precise in Definition 2.8).

**Definition 2.8** (Event and event count)**.** Fix a measurable *Poincaré section* $\Sigma \subset \Omega$ with $\mu(\Sigma) > 0$ that the flow crosses transversally and recurrently (its existence is guaranteed by Theorem 2.3 for any positive-measure $\Sigma$). An *event* at time $t$ is a transversal crossing of $\Sigma$ by the trajectory $\mu(\cdot)$ at $t$. The *event count* is

$$N(t) := \#\{\, s \in (0, t] : \mu(s) \in \Sigma \text{ is a transversal crossing} \,\}. \tag{2}$$

*Remark* 2.9 (Why a count, and why it is time). Definition 2.8 formalises the operational fact that elapsed time is always read from a counter: a pendulum counts swings, a quartz watch counts crystal vibrations, the SI second is a stipulated number of caesium hyperfine cycles. A "cycle" here is a topological event—the trajectory's return to a section—and as such it presupposes no prior notion of duration; duration is what one *defines* by promoting one counter to a standard and stipulating how many of its crossings make a unit. We use $N$ as the primitive temporal quantity and recover the continuous parameter $t$ as the rescaled limit of $N$ for a high-frequency reference. This is why "time reversal" will turn out to be a statement about the flow's solution set (Remark 2.6) and not about $N$.

**Principle 2.10** (The conflation)**.** Loschmidt's reversibility intuition rests on the tacit identification

$$\mathcal{O}(t) \;\equiv\; \mu(t) \;\equiv\; N(t):$$

that recovering the observable ($\mathcal{O}(2T) = \mathcal{O}(0)$) is the same as restoring the microstate ($\mu(2T) = \mu(0)$), which is the same as returning to the past ($N(2T) = N(0)$). The three are distinct, and conflating them is what makes "the trajectory retraced" sound like "the system went back."

We will dismantle the conflation in two steps: by proving the count cannot return (Section 3), and by proving that the configurational reversibility which *does* hold is confined to the microstate–observable layer (Section 4).

## 3 The Enumerative Stratum: Monotonicity of the Count

### 3.1 Strict monotonicity

**Theorem 3.1** (Strict monotonicity of the count)**.** *Let $(\Omega, \mu, \phi_t)$ be a bounded, measure-preserving, non-static system, and let $N$ be its event count with respect to a section $\Sigma$ of positive measure. Then $N$ is non-decreasing, and for every $t_1 < t_2$ separated by at least one event, $N(t_2) > N(t_1)$.*

*Proof.* By Definition 2.8, $N(t)$ counts crossings in $(0, t]$, so $t_1 < t_2$ gives $N(t_1) \leq N(t_2)$ (the counted set grows). By Theorem 2.3 applied to $\Sigma$, transversal crossings recur for almost every trajectory; non-staticity ensures the trajectory is not pinned off the section. Hence on any interval $[t_1, t_2]$ containing a crossing, at least one term is added to the count, so $N(t_2) \geq N(t_1) + 1 > N(t_1)$. □

**Corollary 3.2** (Injectivity of the count on time)**.** *For $t_1 \neq t_2$ separated by at least one event, $N(t_1) \neq N(t_2)$.*

### 3.2 The incoherence of decrement

Monotonicity says the count happens not to decrease. The next result says something stronger and more decisive for the paradox: *no operation can decrease it*, because the execution of any operation is itself a counted process.

**Theorem 3.3** (Incoherence of decrement)**.** *Let $(\Omega, \mu, \phi_t)$ be bounded, measure-preserving, and non-static. There is no physical operation $\Theta$—understood as a process occupying a positive interval of the flow—whose execution leaves the system with $N$ strictly smaller than before $\Theta$ began.*

*Proof.* Suppose $\Theta$ begins at time $a$ and ends at time $b > a$ (positive duration is definitional: an operation that occupies no interval of the flow performs nothing). During $[a, b]$ the trajectory evolves under $\phi_t$, and by Theorem 3.1 it accrues at least the crossings of $\Sigma$ that occur in $[a, b]$, so $N(b) \geq N(a)$, with strict inequality whenever $[a, b]$ contains a crossing. The proposed decrement requires $N(b) < N(a)$, which contradicts this inequality. The only escape would be a $\Theta$ with $b \leq a$, i.e. one occupying no forward interval, which performs no operation. Hence no decrementing operation exists. □

**Corollary 3.4** (The reversal map does not act on the count)**.** *The time-reversal involution $R$ of Definition 2.5 does not decrement $N$. Applying $R$ is a substitution within the solution set of the equations of motion (Remark 2.6); it is not a process occupying a forward interval of the flow, and therefore acts on the configurational data $(q, p)$, not on $N$. Any* physical *realisation of velocity reversal—an apparatus that flips momenta—is itself an operation with positive duration and so, by Theorem 3.3, increments $N$.*

*Remark* 3.5 (The arrow of time as a counting fact). Theorem 3.3 locates the arrow of time not in a statistical tendency layered over reversible foundations, but in the structure of counting itself. The count cannot decrease because counting is what elapsed time *is* (Remark 2.9), and any attempt to "go back" is a further act of counting forward. This is a sharper statement than improbability: the reversed evolution is not rare, it is enumeratively foreclosed. We emphasise that nothing here forbids the *configuration* from retracing its path (Section 4 shows it may); what is foreclosed is the identification of that retracing with a decrement of the count.

### 3.3 Visited-cell entropy

We now connect the count to entropy, in a way that makes the second law a corollary of monotonicity rather than of probability. Fix a countable measurable partition $\mathcal{P} = \{C_1, C_2, \dots\}$ of $\Omega$ into cells of positive measure—the abstract image of a finite-resolution measurement, which can resolve only finitely or countably many distinct outcomes (Cover and Thomas, 2006; Shannon, 1948).

**Definition 3.6** (Visited set and visited-cell entropy)**.** The *visited set* of the trajectory up to time $t$ is the family of cells it has entered,

$$\mathcal{V}(t) := \{\, C \in \mathcal{P} : \mu(s) \in C \text{ for some } s \in [0, t] \,\}. \tag{3}$$

The *visited-cell entropy* is $S(t) := k_{\mathrm{B}} \ln |\mathcal{V}(t)|$.

**Theorem 3.7** (Monotone entropy)**.** *For a bounded, measure-preserving, non-static system, $\mathcal{V}(t_1) \subseteq \mathcal{V}(t_2)$ for $t_1 \leq t_2$, with strict inclusion on a set of positive measure of times. Consequently $S(t)$ is non-decreasing, and strictly increasing whenever the trajectory enters a previously unvisited cell.*

*Proof.* Inclusion is immediate from (3): a cell entered by time $t_1$ has been entered by any later $t_2$, and the membership predicate is never retracted—a cell, once entered, has been entered. Strictness: by Theorem 2.3 the trajectory visits arbitrary neighbourhoods of its starting point, and by non-staticity it is not confined to a single cell; hence on a positive-measure set of times it enters cells not in $\mathcal{V}(t_1)$. Each such entry enlarges $\mathcal{V}$, so $|\mathcal{V}(t_2)| > |\mathcal{V}(t_1)|$ and $S(t_2) > S(t_1)$. □

**Corollary 3.8** (Second law)**.** *The visited-cell entropy of a bounded, measure-preserving, non-static isolated system is a non-decreasing function of time.*

*Remark* 3.9 (Why this is not Boltzmann's argument)*.* The monotonicity of Theorem 3.7 is not probabilistic. It does not say that the visited set typically grows, or grows with high probability; it says the visited set cannot shrink, because set membership is monotone under union. The content is combinatorial—a record that is appended to and never erased—not statistical. Boltzmann's $H$-theorem (Boltzmann, 1877), which quantifies the *rate* of growth under a molecular-chaos assumption, is a statistical refinement of this combinatorial skeleton, not its foundation. The reason velocity reversal cannot decrease $S$ is now visible in advance: reversal rearranges the microstate, but the set of cells *already visited* is a fact about the past trajectory, and the past trajectory is not undone by anything done at the present microstate.

*Remark* 3.10 (Identity as the unvisited complement)*.* The complement $\mathcal{P} \setminus \mathcal{V}(t)$—the cells the trajectory has *not* entered—is a second, equivalent encoding of the visited set, and its measure shrinks monotonically as $\mathcal{V}$ grows. There is a sense in which a trajectory's identity at time $t$ is carried as much by what it has excluded as by where it is: two trajectories with the same current cell but different histories are distinguished precisely by their visited sets, hence by their complements. This "identity by exclusion" will reappear in the spin echo (Section 10), where the irreversible part of the dynamics is exactly the part that has committed information to the environment and so cannot be read back from the system alone.

# 4 The Configurational Stratum and Its Reversibility

We now turn to the stratum on which reversibility genuinely holds, and define it with enough structure to host virtual sub-states in Section 5.

## 4.1 Composition laws and decompositions

**Definition 4.1** (Composition law)**.** Let $V$ be a set of *values* (e.g. a coordinate range, a momentum, a field amplitude) equipped with a partial operation $\boxplus \colon V^k \rightharpoonup V$ for each arity $k \geq 1$, the *composition law*. A *decomposition* of a value $v \in V$ of arity $k$ is a tuple $(v_1, \ldots, v_k) \in V^k$ with $v_1 \boxplus \cdots \boxplus v_k = v$. We write $\mathrm{Dec}_k(v)$ for the set of arity-$k$ decompositions of $v$.

The composition law abstracts the way a physical quantity is assembled from parts: the way a net coordinate is a mean of sub-coordinates, a total momentum a sum of constituent momenta, a measured amplitude a sum over contributing channels. The crucial feature, exploited throughout, is that $\mathrm{Dec}_k(v)$ is in general very large, and that recomposition discards which decomposition was used.

**Definition 4.2** (Configurational state and configurational stratum)**.** Fix a value space $V$, a composition law $\boxplus$, and a *recomposition observable* $\mathcal{O}_V\colon V \to V$ that returns the value a decomposition recomposes to ($\mathcal{O}_V(v) = v$ on values; on a decomposition $(v_1, \dots, v_k)$ it returns $v_1 \boxplus \cdots \boxplus v_k$). The *configurational state* of the system at time $t$ is the pair $(\mu(t), \mathcal{O}_V)$. The *configurational stratum* is the space of all decompositions of all values, quotiented by $\mathcal{O}_V$: two decompositions are configurationally equivalent iff they recompose to the same value.

*Remark* 4.3 (What lives where)*.* The microstate $\mu(t)$ and any coarse observable $\mathcal{O}(\mu(t))$ live on the configurational stratum: they are functions of the instantaneous configuration. The count $N(t)$ does not: it is a function of the *history* of crossings (Definition 2.8), not of the present configuration. This is the formal version of the slogan that the present configuration carries no timestamp—it cannot, since by recurrence (Theorem 2.3) the same configuration recurs at unboundedly many different counts.

## 4.2 Configurational reversibility

**Theorem 4.4** (Configurational reversibility)**.** *Let $(\Omega, \mu, \phi_t)$ admit a time reversal $R$ (Definition 2.5). Then for every microstate $x$ and every $t$,*

$$\mathcal{O}(R\phi_t R\phi_t(x)) = \mathcal{O}(x) \tag{4}$$

*for every observable $\mathcal{O}$: the configuration is exactly restored by evolving forward, reversing, evolving forward again, and reversing once more. In particular, on the configurational stratum the dynamics is reversible.*

*Proof.* By (1), $\phi_{-t} = R\phi_t R$, so $R\phi_t R\phi_t = \phi_{-t}\phi_t = \phi_0 = \mathrm{id}$. Applying any $\mathcal{O}$ to $\mathrm{id}(x) = x$ gives (4). □

*Remark* 4.5 (The two strata do not interfere)*.* Theorem 4.4 and Theorem 3.3 are both true and not in tension: the first says the configuration returns to $x$ under the composite map; the second says that performing that composite map is a process with positive duration that increments $N$. The composite map restores $\mu$ and $\mathcal{O}$ but advances $N$. The system is in the same configuration at a strictly later count. This is the precise content of "the trajectory retraced but the system did not go back," and it is the entire resolution in miniature; the remaining sections enrich the configurational stratum with virtual sub-states and supply physical realisations.

**Theorem 4.6** (Genuine distinctness of the three quantities)**.** *The microstate $\mu$, the coarse observable $\mathcal{O}$, and the count $N$ are logically independent in the following sense.*

*(i) $\mathcal{O}$ may return to a past value while $\mu$ does not: a coarse observable is many-to-one, so $\mathcal{O}(t_2) = \mathcal{O}(t_1)$ does not imply $\mu(t_2) = \mu(t_1)$.*

*(ii) $\mu$ may return (arbitrarily close) to a past value while $N$ does not: by recurrence (Theorem 2.3) the microstate revisits any neighbourhood, but by monotonicity (Theorem 3.1) the count has strictly increased meanwhile.*

*(iii) $N$ never returns to a past value (Theorem 3.3).*

*Proof.* (i) is the definition of a coarse observable (Definition 2.7). (ii): at a recurrence time $t_2$ with $\mu(t_2)$ near $\mu(t_1)$, Theorem 3.1 gives $N(t_2) > N(t_1)$ because crossings accrued in the interim. (iii) is Theorem 3.3. □

*Remark* 4.7 (Dissolution of the recurrence objection)*.* Theorem 4.6(ii) settles Zermelo's recurrence objection to Boltzmann (Boltzmann, 1877; Zermelo, 1896): a Poincaré recurrence is a return of $\mu$, not of $N$, and it is $N$—through the visited set $\mathcal{V}$ of Theorem 3.7—that carries entropy. The microstate may recur a googol times; the visited set only grows.

# 5 Virtual Sub-States

We now show that the configurational stratum is far larger than the microstate, because the decomposition space $\mathrm{Dec}_k(v)$ contains elements whose components are not themselves admissible configurations. These are the virtual sub-states. They are the structure that makes the reversibility of the configurational stratum *do work*—in the backward completion of Section 6 and in the physical realisations of Sections 8–9.

## 5.1 Admissibility and virtuality

**Definition 5.1** (Admissible region)**.** Within the value space $V$, fix a measurable *admissible region* $A \subseteq V$: the set of values that correspond to a physically realisable configuration (a coordinate inside the unit cell, a four-momentum on the mass shell, a partition coordinate inside $[0, 1]$). A value in $A$ is *on-shell*; a value in $V \setminus A$ is *off-shell.*

**Definition 5.2** (Virtual sub-state)**.** Let $v \in A$ be an admissible value and $(v_1, \dots, v_k) \in \mathrm{Dec}_k(v)$ a decomposition. The decomposition is a *virtual sub-state* of $v$ if at least one component $v_i$ is off-shell, $v_i \in V \setminus A$. It is a *physical* (on-shell) decomposition if every component is admissible.

The defining tension is that a virtual sub-state recomposes to an admissible value through inadmissible parts. The composition law permits this precisely because $\mathcal{O}_V$ records only the composite, discarding the parts.

## 5.2 Existence and abundance

We make the abundance precise for the two composition laws that recur in the physical realisations: the mean (Section 9) and the sum under a linear conservation constraint (Section 8). Take $V = \mathbb{R}^m$ with the arity-$k$ mean $\boxplus(v_1, \dots, v_k) = \frac{1}{k}\sum_i v_i$ and admissible region $A = [0, 1]^m$.

**Theorem 5.3** (Existence of virtual sub-states)**.** *Let $v \in (0, 1)^m$ and $k \geq 2$. Then $\mathrm{Dec}_k(v)$ contains virtual sub-states; indeed the set of $(v_1, \dots, v_k)$ with $\frac{1}{k}\sum_i v_i = v$ and at least one $v_i \notin [0, 1]^m$ has positive Lebesgue measure within the constraint affine subspace $\{\, (v_1, \dots, v_k) : \frac{1}{k}\sum_i v_i = v \,\}$.*

*Proof.* The constraint set is an affine subspace of $(\mathbb{R}^m)^k$ of dimension $m(k-1)$, carrying a natural Lebesgue measure. The on-shell decompositions are its intersection with the box $([0, 1]^m)^k$, a bounded convex set of finite measure. The full constraint subspace is unbounded (one may send $v_1 \to +\infty$ along any axis while compensating with $v_2 \to -\infty$, holding the mean fixed), hence of infinite measure. The off-shell decompositions are the complement of a finite-measure set in an infinite-measure space and therefore have positive—indeed infinite—measure. Concretely, for $m = 1$, $v \in (0, 1)$, $k = 2$, the family $(v + c,\ v - c)$ for $c > 1 - v$ has first component $> 1$, hence off-shell, and is a one-parameter family of virtual sub-states. □

**Theorem 5.4** (The composite constrains no single part)**.** *For the mean law on $V = \mathbb{R}^m$ with admissible region $[0, 1]^m$, fix any admissible composite $v \in [0, 1]^m$, any arity $k \geq 2$, and any prescribed values $w_1, \dots, w_{k-1} \in \mathbb{R}^m$ for the first $k - 1$ parts (however far off-shell). Then there is a unique $k$-th part $w_k = k\,v - \sum_{i<k} w_i$ making $(w_1, \dots, w_k)$ a decomposition of $v$.*

*Proof.* The mean constraint $\frac{1}{k}\sum_i w_i = v$ is a single affine equation in $w_k$ with solution $w_k = k\,v - \sum_{i<k} w_i$, which exists and is unique. □

*Remark* 5.5 (Local infeasibility, global feasibility)*.* Theorem 5.4 is the formal content of the slogan that a globally admissible quantity imposes *no constraint whatever* on any proper subset of its parts: one may fix all but one part arbitrarily, and the remaining part absorbs the constraint. The components may be wildly off-shell—“locally infeasible”—while their composite is on-shell and “globally feasible.” This is exactly the freedom that the reversibility of the configurational stratum supplies and that the count knows nothing about. It is also, as the next two physical sections show, not a formal curiosity but the operative mechanism behind off-shell internal lines in field theory and phase-coherent sub-components in spectrometry.

*Remark* 5.6 (Unobservability by recomposition). Because $\mathcal{O}_V$ records only the composite, two distinct decompositions of the same admissible value are indistinguishable by any measurement of that value. We will call this *path opacity*: the recomposed quantity contains no record of which decomposition produced it. Path opacity is what makes virtual sub-states unobservable in isolation while remaining indispensable to the composite—the precise statement, in field theory, that single Feynman diagrams are not observable but their coherent sum is (Section 8).

# 6 The Forward/Backward Asymmetry

We can now prove the keystone. Virtual sub-states are available for one purpose and not its mirror image: they may be used to *complete* a computation toward a fixed admissible endpoint, but not to *construct* a trajectory forward step by step. This asymmetry is the configurational shadow of the enumerative monotonicity of Section 3, and it is what blocks any attempt to use the freedom of the configurational stratum to run the dynamics backward.

## 6.1 Forward construction versus backward completion

**Definition 6.1** (Forward construction)**.** A *forward construction* produces a sequence of configurations $v^{(0)}, v^{(1)}, \ldots, v^{(n)}$ in which each $v^{(j+1)}$ is obtained from $v^{(j)}$ by a step that must itself be an admissible configuration: each intermediate $v^{(j)} \in A$.

**Definition 6.2** (Backward completion)**.** A *backward completion* is given a fixed admissible endpoint $v^{(n)} = v^* \in A$ and produces, for each level $j$, a decomposition of $v^*$'s coordinate at that level, with the only requirement that each decomposition recompose correctly; intermediate components are unconstrained and may be virtual (off-shell).

**Theorem 6.3** (Forward constructions cannot use virtual sub-states)**.** *In a forward construction, no $v^{(j)}$ may be a virtual sub-state: every intermediate is on-shell.*

*Proof.* By Definition 6.1, each $v^{(j)} \in A$. A virtual sub-state has, by Definition 5.2, an off-shell component $v_i \in V \setminus A$; were $v^{(j)}$ itself that component it would lie outside $A$, contradicting admissibility of the intermediate. Hence no intermediate of a forward construction is virtual. □

**Theorem 6.4** (Backward completions may use virtual sub-states)**.** *In a backward completion toward an admissible endpoint $v^*$, the intermediate decompositions may be virtual; moreover, by Theorem 5.3 the set of completions whose intermediates are virtual has positive measure, and by Theorem 5.4 the endpoint constrains no single intermediate component.*

*Proof.* Immediate from Definition 6.2: intermediate components are required only to recompose to $v^*$, not to be admissible. Existence and abundance are Theorems 5.3 and 5.4. □

**Theorem 6.5** (Asymmetry)**.** *The use of virtual sub-states is exclusive to backward completion: a process that admits virtual intermediates is a backward completion toward a fixed endpoint and is not a forward construction; equivalently, restricting a backward completion to on-shell intermediates collapses it to a forward construction.*

*Proof.* By Theorem 6.3 a forward construction admits no virtual intermediate, so any process with a virtual intermediate is not a forward construction. By Theorem 6.4 a backward completion does admit them. Restricting a backward completion to on-shell intermediates imposes $v^{(j)} \in A$ at every level, which is precisely Definition 6.1; the resulting object is a forward construction. □

*Remark* 6.6 (Why the asymmetry is the resolution's keystone). The reason this matters for Loschmidt is the following. The configurational stratum is reversible (Theorem 4.4), and it is rich, because of virtual sub-states (Section 5). One might hope to exploit this richness to drive the system backward—to "complete" the present configuration back into a past one through clever, possibly off-shell, intermediate decompositions. Theorem 6.5 says this hope is misdirected: virtual sub-states complete toward a *fixed* endpoint; they do not run a trajectory in either temporal direction. The endpoint they complete toward is whatever admissible configuration the system is *already at*. The freedom is entirely internal to a single count—it is freedom in how the present configuration may be analysed, not freedom to change which count one is at. The count advances regardless (Theorem 3.3). Thus the richest available reversible structure is powerless against the enumerative arrow: it operates within a count, never across counts.

### 6.2 Cost of the on-shell restriction

The asymmetry has a quantitative face: completions that forgo virtual sub-states lose the efficiency the virtual ones provide. We record the combinatorial fact for the ternary case, which recurs physically in Section 9.

**Proposition 6.7** (Decomposition multiplicity). *The number of ordered decompositions (compositions) of a positive integer $n$ into a sequence of positive parts is $2^{n-1}$; the number of such compositions whose parts are additionally labelled by one of $d$ channels is $d\,(d+1)^{n-1}$.*

*Proof.* A composition of $n$ corresponds bijectively to a choice of a subset of the $n-1$ gaps between $n$ unit dots at which to place a divider, giving $2^{n-1}$ (Stanley, 2011). Labelling each of the $k$ parts of a $k$-part composition by one of $d$ channels multiplies the count for that $k$ by $d^k$; summing over $k$, $\sum_{k=1}^{n}\binom{n-1}{k-1}d^k = d\sum_{j=0}^{n-1}\binom{n-1}{j}d^j = d(1+d)^{n-1}$ by the binomial theorem. □

*Remark* 6.8. The exponential multiplicity of Proposition 6.7 is the size of the decomposition space a backward completion may range over; restricting to on-shell intermediates prunes it to the (generally much smaller) admissible subset. In the embedding of Section 9 this is the difference between $\mathcal{O}(\log N)$ and $\mathcal{O}(N)$ work to resolve a feature at depth $\log_d N$. The same multiplicity underlies the abundance of off-shell decompositions in Theorem 5.3.

## 7 The Resolution of Loschmidt's Paradox

We assemble the resolution.

**Theorem 7.1** (Velocity reversal cannot decrease entropy). *Let $(\Omega, \mu, \phi_t)$ be bounded, measure-preserving, non-static, and admit a time reversal $R$. Let the system evolve from $t_0$ to $T$ and let an apparatus then perform a physical velocity reversal. Then:*

*(i) the configuration thereafter retraces, and any coarse observable can return to a past value (Theorem 4.4);*

*(ii) the event count strictly increases throughout, including across the reversal operation (Theorems 3.1, 3.3, Corollary 3.4);*

*(iii) the visited-cell entropy $S$ does not decrease, because the visited set is a monotone function of the trajectory's history and the reversal acts only on the present configuration (Theorem 3.7, Remark 3.9).*

*Consequently velocity reversal restores configuration without restoring the past: the system is in a retraced configuration at a strictly later count and strictly larger (or equal) entropy. No decrease of total entropy occurs.*

| Account | Why reversal is not seen | Residual commitment |
|---|---|---|
| Boltzmann statistical (Boltzmann, 1877) | Reversed microstates are astronomically rare. | Reversal possible, merely improbable. |
| Gibbs coarse-graining (Gibbs, 1902) | Coarse-grained entropy grows; fine-grained is constant. | Irreversibility is observer-relative. |
| Information-theoretic (Jaynes, 1957) | Entropy is ignorance of the microstate. | Irreversibility is epistemic. |
| Cosmological boundary (Penrose, 1989) | A low-entropy initial condition sets the gradient. | Explains the global gradient, not local reversal. |
| Decoherence (Zurek, 2003) | Entanglement with the environment destroys phase relations. | Closed-system reversal granted as coherent. |
| **This work** | Configuration retraces; the count and its visited-set entropy do not decrease, because reversal acts only on the present configuration. | None: reversal is coherent configurationally and absent enumeratively (Thm. 3.3). |

Table 1: The standard resolutions accept the reversibility presupposition and explain its consequences away; the present account separates the configurational and enumerative strata and locates the arrow in the latter.

*Proof.* (i) is Theorem 4.4. (ii): the pre-reversal evolution increments $N$ by Theorem 3.1; the reversal apparatus is a physical operation of positive duration, so by Theorem 3.3 it too increments $N$, and the mathematical involution $R$ does not act on $N$ by Corollary 3.4; the post-reversal evolution increments $N$ again. (iii): by Theorem 3.7, $\mathcal{V}$ only grows; the reversal changes $\mu$ but not the family of cells already entered, so $S$ cannot fall. The final clause combines (i)–(iii). □

**Corollary 7.2** (Dissolution of the paradox)**.** *Loschmidt's reversal is neither improbable nor forbidden: the configuration genuinely retraces. What does not happen is the identification of that retracing with a return to the past, because the count—the operational content of elapsed time—strictly increases and carries the monotone visited-set entropy with it. The paradox is the conflation of Principle 2.10; separating the strata dissolves it.*

*Remark* 7.3 (Comparison with the standard resolutions). Table 1 situates the present account. Each standard resolution grants the reversibility presupposition—that time reversal is a coherent, in-principle-achievable operation—and then explains why its consequences are not seen. The present account denies the presupposition at its root: the operation is coherent on the configurational stratum (where it changes nothing about the arrow) and has no realisation on the enumerative stratum (where the arrow lives). The denial is not an extra assumption; it is Theorem 3.3, which follows from boundedness, measure preservation, and the definition of an operation as a process of positive duration.

# 8 Physical Realisation I: Off-Shell States in Quantum Field Theory

The abstract virtual sub-state of Section 5 has a precise, long-established physical instance: the off-shell internal line of a Feynman diagram. We develop the correspondence and show that path opacity (Remark 5.6) is exactly the statement that the sum over diagrams, not the individual diagram, is observable.

## 8.1 The off-shell line as a virtual sub-state

In perturbative quantum field theory (Itzykson and Zuber, 1980; Peskin and Schroeder, 1995; Weinberg, 1995), a scattering amplitude for external particles of momenta $p_1, \ldots, p_n$ is computed as a sum over Feynman diagrams (Dyson, 1949; Feynman, 1949). External lines carry on-shell momenta, $p_i^2 = m_i^2$ (in units $\hbar = c = 1$); internal lines carry momenta $q$ constrained only by momentum conservation at each vertex, and generically $q^2 \neq m^2$—they are *off-shell*, or virtual.

**Proposition 8.1** (Internal lines are virtual sub-states)**.** *Take the value space $V$ to be four-momentum space $\mathbb{R}^{1,3}$, the admissible region $A = \{q : q^2 = m^2\}$ the mass shell, and the composition law the sum constrained by conservation at each vertex. Then the internal lines of any diagram contributing to a fixed on-shell external configuration form a virtual sub-state of that configuration in the sense of Definition 5.2: their momenta recompose (via conservation) to the on-shell external data, while each internal momentum is generically off-shell.*

*Proof.* Momentum conservation at the vertices is a system of linear constraints expressing the external momenta as a signed sum of internal and external line momenta; this is the composition law $\boxplus$ specialised to the constrained sum. The external data $\{p_i\}$ lie in $A$ by construction (asymptotic states are on-shell). For a loop diagram the loop momentum is integrated over all of $\mathbb{R}^{1,3}$, the overwhelming part of which has $q^2 \neq m^2$; for a tree internal line the momentum is fixed by conservation and generically off-shell. Hence the internal configuration is a decomposition of the external data with off-shell components: a virtual sub-state. □

*Remark* 8.2 (The propagator weights the off-shell excursion). The Feynman propagator $i/(q^2 - m^2 + i\varepsilon)$ assigns each internal line a weight that is largest near the mass shell and falls off as the line goes further off-shell; the $i\varepsilon$ prescription fixes the contour. The off-shell excursion is thus not forbidden but weighted—exactly as in the abstract picture, where off-shell decompositions are permitted and abundant (Theorem 5.3) but the dynamics supplies a weighting over them. For an unstable particle the propagator denominator $q^2 - m^2 + im\Gamma$ carries an imaginary part $im\Gamma$ encoding the decay rate; the real part is the off-shell distance and the imaginary part the rate of committed transition.

## 8.2 Diagrams as backward completions; path opacity

**Proposition 8.3** (A diagram is a backward completion)**.** *A Feynman diagram contributing to a fixed on-shell external configuration $v^* = \{p_i\}$ is a backward completion (Definition 6.2) of $v^*$: the endpoint (external lines) is fixed and on-shell, the intermediate components (internal lines) are unconstrained except by recomposition (conservation), and they may be virtual.*

*Proof.* The external data are fixed and admissible; the internal data satisfy only the recomposition constraint and are generically off-shell (Proposition 8.1). This is exactly Definition 6.2. □

**Theorem 8.4** (Path opacity and the sum over diagrams)**.** *Two diagrams $\Gamma_1, \Gamma_2$ with the same external lines but different internal topologies are distinct backward completions of the same on-shell endpoint. No measurement performed on the external (asymptotic) states distinguishes them; only their coherent sum $\mathcal{M} = \sum_\Gamma \mathcal{M}_\Gamma$ enters the observable $|\mathcal{M}|^2$.*

*Proof.* By Proposition 8.3 each $\Gamma_a$ is a backward completion of the same endpoint $v^*$; by path opacity (Remark 5.6), the recomposed quantity—here the $S$-matrix element evaluated on the external momenta—records only the composite, not which internal decomposition produced it. The observable transition probability is $|\mathcal{M}|^2 = |\sum_\Gamma \mathcal{M}_\Gamma|^2$, which contains interference cross terms $\mathcal{M}^*_{\Gamma_1}\mathcal{M}_{\Gamma_2}$; the individual $\mathcal{M}_\Gamma$ cannot be recovered from $|\mathcal{M}|^2$ without further structure. Hence single diagrams are not observable and their sum is. □

*Remark* 8.5 (Unitarity as recomposition bookkeeping)*.* The optical theorem, $2\,\mathrm{Im}\,\mathcal{M}(p\to p) = \sum_f \int d\Pi_f\, |\mathcal{M}(p\to f)|^2$ (Cutkosky, 1960; Peskin and Schroeder, 1995), is the statement that the imaginary part of the forward amplitude equals the total transition rate summed over final states. In the present language it is a recomposition identity: the rate at which probability flows out of the initial channel through (off-shell) virtual intermediates equals the rate at which it is committed into (on-shell) final channels. The off-shell sub-states are the reversible, path-opaque interior; the committed final-state rate is the enumerative quantity. Unitarity is the conservation law tying the two, the field-theoretic counterpart of the strict bookkeeping between the configurational and enumerative strata.

*Remark* 8.6 (Why virtual particles are not observed)*.* Theorem 8.4 explains the textbook statement that virtual particles cannot be directly observed (Peskin and Schroeder, 1995): a measurement of the external particles cannot determine which internal lines were exchanged, because the observable depends on the recomposed amplitude and not on the decomposition. This is path opacity, not a limitation of apparatus. The off-shell line is a genuine, weighted element of the reversible configurational interior; its unobservability in isolation is the same fact as the unobservability of a single decomposition of a recomposed value (Remark 5.6).

# 9 Physical Realisation II: Phase-Coherent Sub-Components in Fourier Mass Spectrometry

The second realisation is metrological and, unlike the field-theoretic one, directly furnishes a measured off-shell fraction. An ion oscillating in a Fourier-transform mass analyser is a bounded oscillatory system whose transient encodes, in its phase and amplitude structure, decompositions that are partly off-shell.

## 9.1 The oscillating ion as a bounded system

In an Orbitrap analyser (Makarov, 2000) an ion of mass-to-charge ratio $m/z$ executes harmonic axial motion at angular frequency $\omega_z = \sqrt{q\kappa/m}$, with $\kappa$ the field curvature; in a Fourier-transform ion-cyclotron-resonance cell (Comisarow and Marshall, 1974) the relevant frequency is the cyclotron frequency $\omega_c = qB/m$. In both, the ion is a bounded oscillator in the sense of Definition 2.2, the image current it induces is recorded as a time-domain transient, and the mass is inferred from the oscillation frequency—never measured directly, but read from the rate at which the oscillation cycles accrue.

*Remark* 9.1 (Frequency as a count rate)*.* That mass is read from a frequency is the metrological face of Remark 2.9: the analyser is a counter, and the "measurement" of $m/z$ is a comparison of the ion's cycle-accrual rate against the instrument's reference oscillator. The transient is a record of counts, and the event count $N$ of Definition 2.8 is, for this system, literally the number of image-current cycles.

## 9.2 Phase coherence and sub-components

Consider a precursor ion and its dissociation products, all oscillating in the same trap field. Because the products are produced from the precursor's continuous oscillatory trajectory, their phases are coherent with it.

**Proposition 9.2** (Frequency relation of products)**.** *In a fixed Orbitrap field (fixed $\kappa$, fixed charge), a product ion of mass $m_f$ oscillates at*

$$\omega_f = \omega_{\mathrm{prec}}\sqrt{m_{\mathrm{prec}}/m_f}, \tag{5}$$

*where $\omega_{\mathrm{prec}}$ and $m_{\mathrm{prec}}$ are the precursor's frequency and mass.*

*Proof.* From $\omega_z = \sqrt{q\kappa/m}$ with common $q, \kappa$, dividing the product and precursor relations gives $\omega_f/\omega_{\text{prec}} = \sqrt{m_{\text{prec}}/m_f}$. □

*Remark* 9.3 (Sub-components in the transient). By linearity of image-current superposition, a transient containing the precursor and a coherent product carries both a component at $\omega_{\text{prec}}$ and one at $\omega_f$ given by (5), the latter phase-locked to the former. The information that conventional processing discards—phase and the sub-harmonic amplitude structure—is exactly the decomposition data: the record of how the precursor's oscillatory state decomposes into coherent sub-components.

## 9.3 The mean-recovery decomposition and its off-shell fraction

We now exhibit the virtual sub-state structure directly. Normalise the relevant ion parameters (frequency, charge state, polarity, time) onto a common scale so that the physical ion corresponds to an admissible coordinate $v \in [0,1]^m$, and consider decompositions $(v_1, \dots, v_k)$ with mean $v$ (Definition 4.1, mean law).

**Proposition 9.4** (Off-shell sub-components exist and recompose)**.** *For a normalised ion coordinate $v \in (0,1)^m$, the decomposition space contains mean-recovery tuples with off-shell components (some $v_i \notin [0,1]^m$), by Theorem 5.3; their mean recovers the physical coordinate by construction. These off-shell sub-components are the spectrometric virtual sub-states.*

*Proof.* Direct specialisation of Theorem 5.3 with the mean law and admissible region $[0,1]^m$. □

*Remark* 9.5 (A measured off-shell fraction). The content of Proposition 9.4 is empirically accessible: constructing the stacked decomposition of a real ion across its natural parameters and computing the fraction of sub-components falling outside $[0,1]^m$ yields a definite, nonzero off-shell fraction, while the mean of the components remains the physical (admissible) coordinate in every case. A nonzero such fraction is the direct measurement of virtuality: it shows that the reversible configurational interior of a single measured ion contains genuinely off-shell parts whose composite is on-shell. This is the metrological counterpart of the field-theoretic statement (Section 8) that internal lines are off-shell while external lines are on-shell.

*Remark* 9.6 (Tailored-waveform cancellation). Stored-waveform inverse-Fourier-transform excitation (Marshall et al., 1985), which isolates a target ion by ejecting others, is the operational use of off-shell sub-components: the ejection waveform is the inverse transform of a decomposition designed so that the unwanted ions' contributions cancel in recomposition while the target's survives. That such a waveform exists and works is the engineering proof that mean-recovery decompositions with cancelling (and individually off-shell) parts are physically constructible.

## 9.4 The efficient embedding

The ternary refinement underlying the normalised coordinate embeds the discrete partition of resolutions into a continuous metric space, where backward completion is efficient.

**Proposition 9.7** (Fisher embedding and backward efficiency)**.** *The hierarchy of normalised coordinates with ternary refinement embeds into $[0,1]^3$ with the Fisher-information metric $ds^2 = \sum_i (dS^i)^2/(S^i(1-S^i))$, the unique metric invariant under the refinement (a consequence of Čencov's uniqueness theorem (Amari, 2016; Čencov, 1982)). Backward completion of a feature at refinement depth $\log_3 N$ then costs $\mathcal{O}(\log_3 N)$ steps, against $\Omega(N)$ for an on-shell-only search.*

*Proof sketch.* Assign each depth-$n$ cell its base-3 address along each axis and map the address to the corresponding ternary fraction; the induced map is injective with dense image, and the Fisher metric is the unique refinement-invariant Riemannian metric on the resulting simplex coordinates (Amari, 2016; Čencov, 1982). Descending to a leaf at depth $\log_3 N$ is $\log_3 N$ ternary choices; an on-shell-only search among $N$ leaves is $\Omega(N)$, recovering the multiplicity gap of Remark 6.8. □

*Remark* 9.8. Proposition 9.7 is the quantitative form of the asymmetry (Theorem 6.5) in the spectrometric setting: the $\mathcal{O}(\log N)$ efficiency is available only to the backward completion that may pass through off-shell sub-coordinates; restricting to on-shell intermediates forfeits it and returns the cost to $\Omega(N)$.

# 10 The Spin Echo: A Controlled Test of the Reversibility Assumption

The two realisations above instantiate virtual sub-states; the spin echo supplies the experimental verdict on the reversibility presupposition itself. It is the closest laboratory approximation to Loschmidt's operation, performed routinely since 1950 (Hahn, 1950), and its seventy-year refinement exposes an irreducible floor rather than improving reversal.

## 10.1 The echo as configurational recovery

In a spin-echo experiment an ensemble of nuclear spins is tipped into the transverse plane, allowed to dephase under a spread of precession frequencies, and then subjected to a refocusing (180°) pulse that inverts each accumulated phase $\varphi_i \mapsto -\varphi_i$; at twice the dephasing time the ensemble rephases and the macroscopic signal reappears as an echo (Hahn, 1950).

**Proposition 10.1** (The echo recovers configuration, not history). *For the static (inhomogeneous) part of the dephasing, the refocusing pulse restores the observable transverse magnetisation $\mathcal{O}(2\tau) = \mathcal{O}(0)$ while the microstate is never lost (each phase is a determined function of the fixed local offset and the pulse history) and the event count advances monotonically across $[0, 2\tau]$.*

*Proof.* A spin with fixed offset $\Delta\omega_i$ accumulates phase $\Delta\omega_i\tau$ before the pulse; the pulse sends it to $-\Delta\omega_i\tau$, and it accumulates $\Delta\omega_i\tau$ again in $[\tau, 2\tau]$, returning to zero phase at $2\tau$ independently of $\Delta\omega_i$. The vector sum—the observable—is thereby restored: $\mathcal{O}(2\tau) = \mathcal{O}(0)$. Each phase is at every instant a determined function of the fixed offset and the (deterministic) pulse history, so the microstate is never lost; this is configurational reversibility (Theorem 4.4) realised in the laboratory. The spectrometer's reference oscillator crosses its section throughout, so by Theorem 3.1 the count strictly increases. □

*Remark* 10.2 (Echo = retraced configuration at a later count). Proposition 10.1 is Remark 4.5 made experimental: the signal returns (configuration restored) while the count advances (no return to the past). "The signal came back" is not "the system went back." One can point at all three quantities at once on the oscilloscope: the observable that refocused, the microstate that ran forward intact, and the count that advanced and cannot return.

## 10.2 What cannot be refocused, and why

The echo never fully recovers. The fluctuating (homogeneous) part of the dephasing, governed by the transverse relaxation time $T_2$ (Bloembergen et al., 1948), is not refocused, and the echo amplitude is reduced by $e^{-2\tau/T_2}$.

**Proposition 10.3** (Refocusability excludes committed interaction). *A dephasing contribution is refocusable by a 180° pulse if and only if the phase it produces after the pulse is a determined function of the present microstate. A contribution arising from a fluctuating interaction that has committed which-spin information to the environment is not such a function and is not refocusable.*

*Proof.* If the post-pulse phase is determined by the present microstate, the pulse $\varphi \mapsto -\varphi$ pre-compensates it and the contribution refocuses (the static case of Proposition 10.1). Conversely, a fluctuating interaction imprints a phase increment in $[\tau, 2\tau]$ that is statistically independent of the increment in $[0, \tau]$ once which-spin information has passed into the environment; knowing the present microstate of the spin does not determine it, so no operation on the spin alone can pre-compensate it, and the residual randomised phase degrades the realignment. □

**Corollary 10.4** (The irreducible floor is the committed record)**.** *The unrefocusable part of the dephasing is exactly the part that has committed information to the environment—entered new cells of the joint system–environment partition in the sense of Definition 3.6—and the echo deficit* $e^{-2\tau/T_2}$ *is a calibrated readout of that committed record. It is non-negative, monotone in* $\tau$*, and not reduced by any further pulse.*

*Proof.* By Proposition 10.3 the unrefocusable contributions are the committed ones; committing information is entering previously unvisited cells of the joint partition, which by Theorem 3.7 only accumulates. The Gaussian treatment of the fluctuating phase gives a mean residual amplitude $\exp(-\frac{1}{2}\langle\Delta\varphi^2\rangle)$ with $\langle\Delta\varphi^2\rangle \propto 2\tau/T_2$, yielding the stated deficit; its monotonicity in $\tau$ and invariance under further pulses follow from the monotonicity of the committed record. □

### 10.3 The echo hierarchy and the engineering-independent floor

The refocusing idea has been pushed toward ever more complete reversal, and the result is a hierarchy in which better engineering exposes a sharper, engineering-independent floor.

- **Hahn echo** (Hahn, 1950): reverses the static single-spin inhomogeneity; floor at the homogeneous $T_2$, the deficit $e^{-2\tau/T_2}$ of Corollary 10.4.
- **CPMG** (Carr and Purcell, 1954; Meiboom and Gill, 1958): iterates the refocusing pulse, converting slowly varying field components into refocusable (configurational) dephasing before they commit; the decay time lengthens toward, but never beyond, the limit set by fast fluctuations. Packing pulses more densely does not breach the floor.
- **Magic / polarization echo** (Levstein et al., 1998; Rhim et al., 1971): engineers the many-body dipolar evolution to run as though under $-H$ rather than $+H$, the closest laboratory enactment of Loschmidt's operation at the level of the interactions. The measured fidelity decay becomes, in the strongly coupled regime, *independent of the residual imperfection* of the reversal: making the engineering more perfect does not slow the decay (Gorin et al., 2006; Jalabert and Pastawski, 2001; Levstein et al., 1998; Peres, 1984).

*Remark* 10.5 (The verdict)*.* The reversibility presupposition predicts that the barrier to reversal is contingent—improbability, imperfection, openness—and so should recede as control improves. The echo hierarchy falsifies this prediction: the barrier does not recede with control; it is intrinsic, set by the system's own committed dynamics. This is precisely Theorem 3.3 and Corollary 10.4 made experimental. The configurational part refocuses arbitrarily well (Proposition 10.1); the committed part does not refocus at all (Proposition 10.3); and the limit of perfect reversal engineering is not perfect reversal but the bare exposure of the floor. The experiment named after Loschmidt is the experiment that decides against Loschmidt's premise.

## 11 Objections and Replies

**(O1) "Erase the record, and the count is gone."** If one erases the visited-set record—or the observer's memory of it—the states $S_0$ (before evolution) and $S_0'$ (after reversal) might seem indistinguishable, hence the same. The reply is twofold. First, erasure is itself an operation of positive duration and so increments the count (Theorem 3.3); it cannot decrement what it is an instance of. Second, erasure is physically a commitment of information to the environment, the irreversible kind of step of Proposition 10.3 and Corollary 10.4: it enlarges, not shrinks, the visited set of the joint system–environment partition. Erasure removes the ability to *read* the distinction from the system alone; it does not remove the distinction (Bennett, 1982; Landauer, 1961; Szilárd, 1929).

**(O2) "The count depends on the choice of section $\Sigma$."** Different sections yield different counts, but the monotonicity of each (Theorem 3.1) and the incoherence of decrement (Theorem 3.3) hold for every positive-measure section. The arrow is a property shared by all admissible counters, not an artefact of one; this is the counterpart of the fact that all physical clocks agree on the direction of time though not on its rate.

**(O3) "Virtual sub-states are a mere bookkeeping device."** They are a bookkeeping device, and they are physical. The off-shell internal line is indispensable to the computed amplitude yet unobservable in isolation (Theorem 8.4); the spectrometric off-shell fraction is measured (Remark 9.5); the tailored-waveform cancellation is engineered (Remark 9.6). A device with measured fractions and engineered uses is not "mere." What is true is that virtual sub-states are confined to the reversible configurational interior and are silent about the count—which is exactly why they cannot be marshalled to reverse the arrow (Theorem 6.5).

**(O4) "This is just coarse-graining renamed."** It is not. Coarse-graining (Gibbs, 1902) makes irreversibility relative to a choice of smearing and leaves the fine-grained entropy constant by Liouville's theorem. The present account makes no smearing: the visited set $\mathcal{V}$ is defined by which cells the trajectory has *entered*, a fact about the actual path, not about a chosen resolution of the present state. Two trajectories at the same present cell with different histories have different visited sets; coarse-graining of the present state cannot tell them apart, while $\mathcal{V}$ does (Remark 3.10). The irreversibility here is of the record, not of the resolution.

**(O5) "Why is $c$, the mass shell, or $[0, 1]^m$ the right admissible region?"** The resolution does not depend on which admissible region one fixes; it depends only on there *being* one, so that "off-shell" is well defined and the asymmetry of Section 6 has content. The three physical sections supply three concrete admissible regions—the mass shell, the unit cube of normalised ion parameters, and the determined-phase subspace of the spin ensemble—and the same abstract theorems specialise to each.

## 12 Conclusion

Loschmidt's paradox is the appearance of contradiction between the time-reversal symmetry of the microscopic equations and the manifest asymmetry of macroscopic processes. We have argued that the appearance is produced by collapsing two strata of a system's description that obey different logics, and that holding them apart dissolves it without any appeal to improbability, coarse-graining, or special boundary conditions.

The configurational stratum—the instantaneous microstate together with the space of its decompositions—is genuinely reversible (Theorem 4.4), and it is rich, because it contains virtual sub-states: decompositions whose components may be off-shell while their composite is on-shell (Theorem 5.3, Theorem 5.4). The enumerative stratum—the count of completed dynamical events—cannot decrease, because any operation that would decrement it is itself a counted process (Theorem 3.3); it carries a visited-set entropy that is monotone by the combinatorics of an appended record, with the second law as a corollary (Corollary 3.8) and Boltzmann's $H$-theorem as its statistical refinement. The two strata are logically independent (Theorem 4.6), and the freedom of the first cannot reach the second, because virtual sub-states are available only for the backward completion of a fixed endpoint and never for forward construction (Theorem 6.5)—a structural asymmetry that is the configurational shadow of the enumerative arrow. Velocity reversal acts on the configurational stratum and leaves the enumerative one untouched, so it restores configuration without restoring the past (Theorem 7.1).

That virtual sub-states are not a formal contrivance is established by two independent physical realisations and one controlled experiment. In quantum field theory the off-shell internal line is a weighted, path-opaque element of the reversible interior, and the unobservability of single diagrams is the unobservability of single decompositions (Theorem 8.4). In Fourier mass spectrometry a measured fraction of an ion's natural decomposition lies off the physical manifold while its mean recovers the physical coordinate (Remark 9.5). And the nuclear spin echo, the closest laboratory approximation to Loschmidt's own operation, refocuses the configurational part of the dynamics arbitrarily well while exposing an engineering-independent floor exactly where information has been committed (Corollary 10.4, Remark 10.5)—a seventy-year experimental verdict against the assumption that time reversal is a coherent operation on anything but configuration.

The arrow of time, in this account, is neither imposed by a boundary condition nor purchased with a probability: it is the monotonicity of counting, and the reason the richest reversible structure we possess cannot overturn it is that this structure lives entirely within a single count and is constitutively silent about which count one is at.

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